OPEN HARDWARE

DIY AI GPU SERVER

20×20 aluminium extrusion frame · 5–6 × RTX 4090 · 19″ / 14U



Two people needed.

Assembly guide · 2026-10-29

Important Read before you start

From step 4 on the frame is bulky. Build it with two people.

Forgot a T-nut?
I-type slot 5: T-nuts can **not** be added later. Slide in **all** of them before joining (step 2).

ESD
From step 10 (mainboard) ground an ESD strap to the frame.

Mains power
Plug in the power cords last. Never re-plug anything while powered.

Square
Check every corner with a try square, **then** tighten.

Kinking SlimSAS
Bend radius ≥ 25 mm. The first 45 mm at the plug are rigid – do not bend.

Two cables stick out on the left. The SlimSAS cables of GPU1 run over the mid frame and down the outside of the left side (approx. 15 mm). Leave at least 17 mm on the left in the rack; no closed left side panel.

Airflow. All graphics cards face their fans to the **left** (−X). A gap of 16,8 mm stays in front of each card. Keep it clear.

Dimensions

Outside (W × D × H)	430 × 500 × 621 mm (incl. 50 mm feet)
Rack	19″, 14U (622.3 mm), stands on rails or a shelf; rack ears optional
Frame weight	approx. 8 kg (extrusions + brackets)
Graphics cards	5 × RTX 4090 (305 × 61 × 125 mm), optional 6th card on the board level
Power supplies	2 × Silverstone ZEUS 1650R, upright, inlet down

Tools

2.5

Hex key 2.5 mm
all DIN 7984 M4 screws (brackets, holders, fans)

PH2

Phillips PH2
PSUs, slot brackets, cooler mount

PH1

Phillips PH1 / Torx T8
M3 on board and risers, M.2

°C

Soldering iron
M3 heat-set inserts in the standoffs (approx. 220 °C)

90°

Try square
check every corner before tightening

mm

Tape measure / calipers
diagonals, distances

RM

Rubber mallet
gently align extrusions



Side cutters
trim cable ties

ESD

ESD wrist strap
from mainboard installation on

Consumables

Qty	Material	Used for
40×	Heat-set insert M3 (Ø4, 5.7 mm)	16 standoffs 8×10, 24 standoffs 8×18
40×	Socket head screw M3×6	mainboard, M.2 riser, GPU risers
50×	Cable ties 2.5 × 100 / hook-and-loop	SlimSAS and power cables
1×	Thermal paste (included with NH-U14S)	CPU
1×	Isopropyl alcohol + lint-free cloth	CPU, cooler
2×	IEC C13 power cord	PSUs
~0.6 kg	PETG filament	all printed parts (STL/3MF in print/)

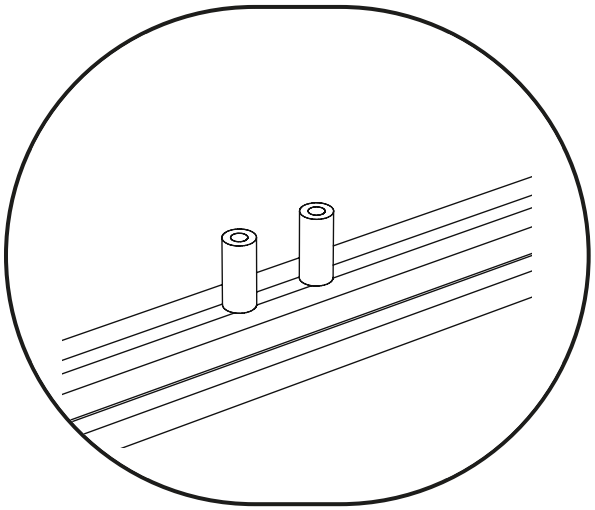
Parts

A **21×** **Extrusion 390 mm** 390 mm cut, 20x20 I-type slot 5	B **8×** **Extrusion 460 mm** 460 mm cut	C **4×** **Extrusion 621 mm** 621 mm cut (571 + 50 mm foot)	W **58×** **Bracket + M4 set** 20x20x20, incl. 2× M4×8 + 2× T-nut	D **16×** **Standoff 8×10** board + M.2 riser standoffs
E **24×** **Standoff 8×18** GPU riser standoffs · 4 optional	F **4×** **PSU holder** PETG, 4 different	G **3×** **Bracket comb** PETG, 3 segments	H **5×** **GPU rest** PETG	1 **2×** **ZEUS 1650R** Silverstone ZEUS 1650R
2 **1×** **Mainboard** ASRock Rack ROMED16QM3	3 **1×** **NH-U14S** Noctua NH-U14S TR4-SP3	4 **8×** **RDIMM** DDR4 RDIMM	5 **1×** **NIC 40G** low profile, PCIE8	6 **1×** **Quad M.2** quad M.2 card + riser
7 **6×** **C-Payne riser** C-Payne x8/x16 BOTTOM rev. 2 · 1 optional	8 **6×** **RTX 4090** 305 × 61 × 125 mm · 1 optional	S **14×** **SlimSAS 8i 500** SFF-8654, 500 mm, round · 2 optional	P **14×** **PSU cable** PSU cables + IEC plugs	9 **3×** **Fan 120** 120 × 120 × 25 mm
10 **2×** **Rack ear** optional	11 **1×** **GPU6 holder** optional	K **8×** **End cap** 20 I-type slot 5	N **188×** **T-nut M4** I-type slot 5 with web (116 come with the brackets)	M4 **46×** **M4 screws** DIN 7984: 12× M4×30, 26× M4×12, 8× M4×10

1 Prepare the printed parts

(D) **16×** Standoff 8×10

(E) **20×** Standoff 8×18



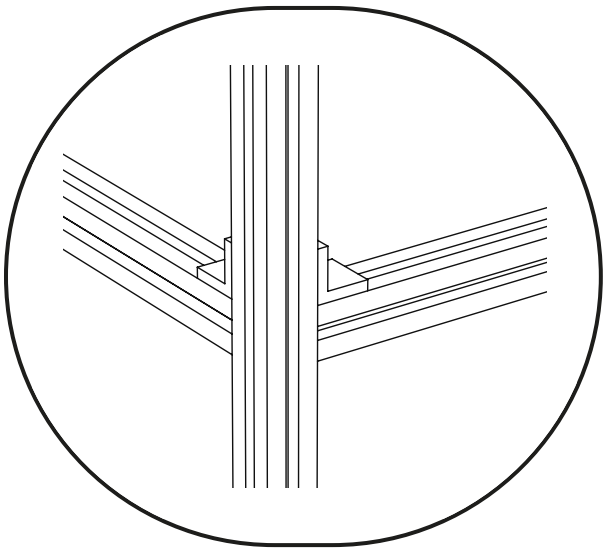
Press an M3 heat-set insert straight into the top hole of every standoff with the soldering iron (approx. 220 °C) until flush.

flush, straight ✓

tilted / proud ✗

2 Slide in the T-nuts

(N) **188×** T-nut M4

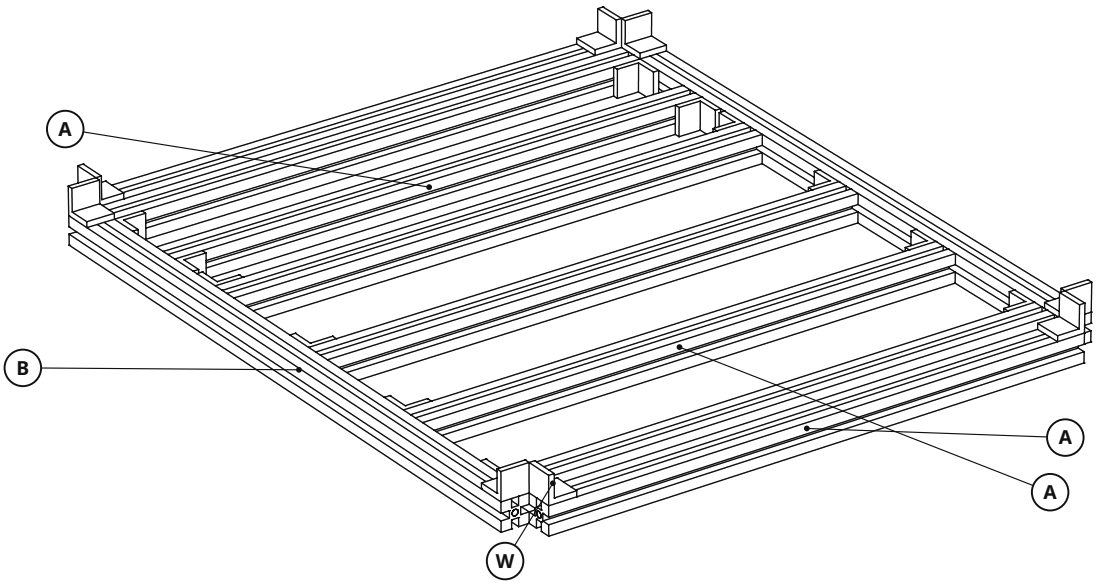


Before every joint slide the T-nuts into the slot – **2** per bracket (one per extrusion), plus those for standoffs, holders and fans.

Extrusion	Qty
Base front	**2**
Base left	**8**
Base rear	**2**
Base right	**8**
Board rail 1	**5**
Board rail 2	**5**
Board rail 3	**7**
Board rail 4	**7**
Front fan rail lower	**8**
Front fan rail upper	**8**
GPU bracket rail	**2**
GPU front support	**12**
GPU side rail left	**3**
GPU side rail right	**3**
Mid frame front	**2**
Mid frame left	**5**
Mid frame rear	**2**
Mid frame right	**5**
NIC rail	**2**
PSU rail 1	**10**
PSU rail 2	**2**
Post S1	**10**
Post S2	**10**
Post S3	**9**
Post S4	**9**
Rear rail lower	**2**
Rear rail upper	**2**
Riser rail 1	**12**
Riser rail 2	**18**
Top front	**2**
Top left	**2**
Top rear	**2**
Top right	**2**

3 Base frame

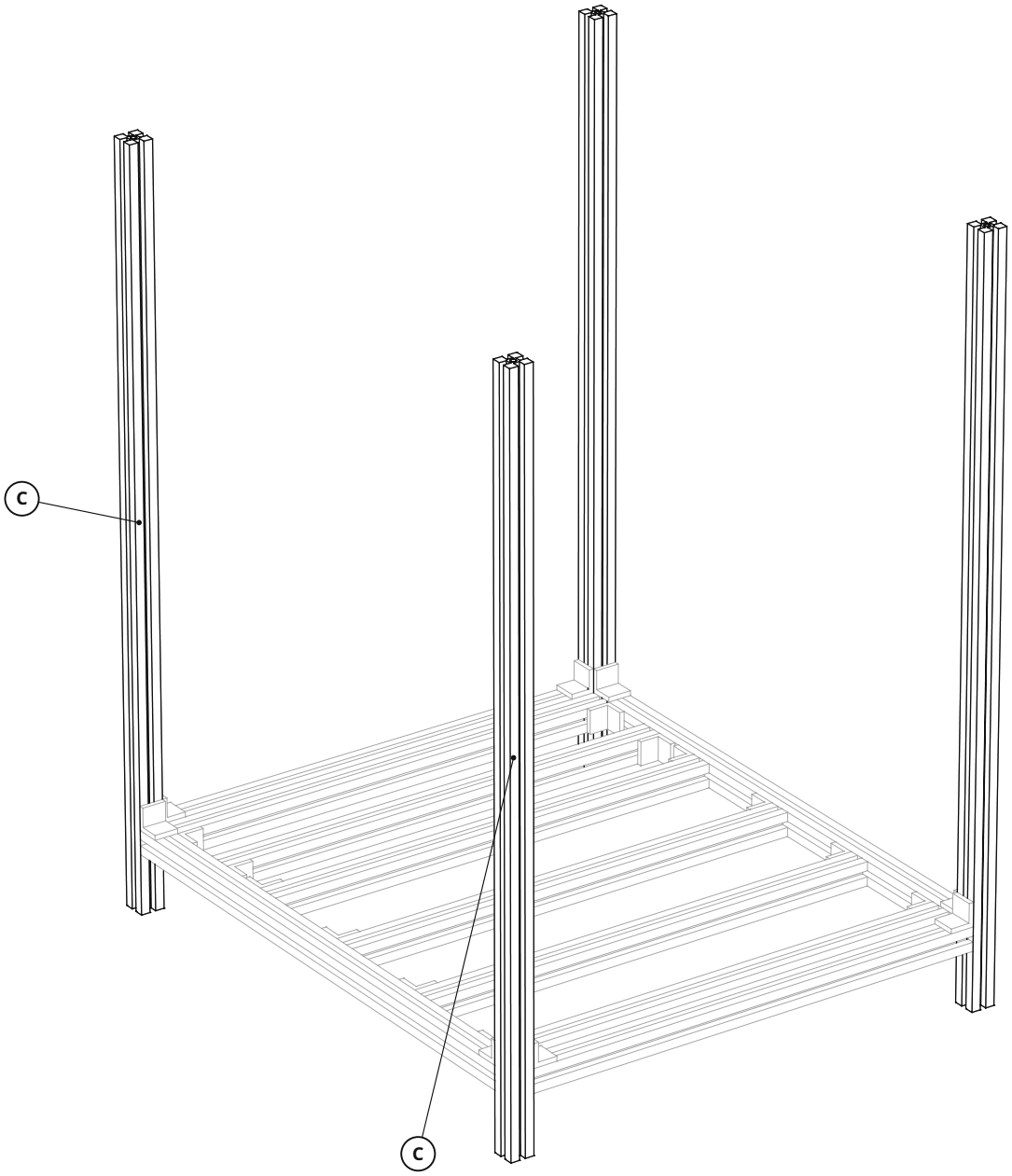
A	**8×**	Extrusion 390 mm	B	**2×**	Extrusion 460 mm
W	**20×**	Bracket + M4 set			



! Build on a flat surface. Tighten the brackets lightly, measure both diagonals (equal length), then tighten fully.

4 Corner posts with feet

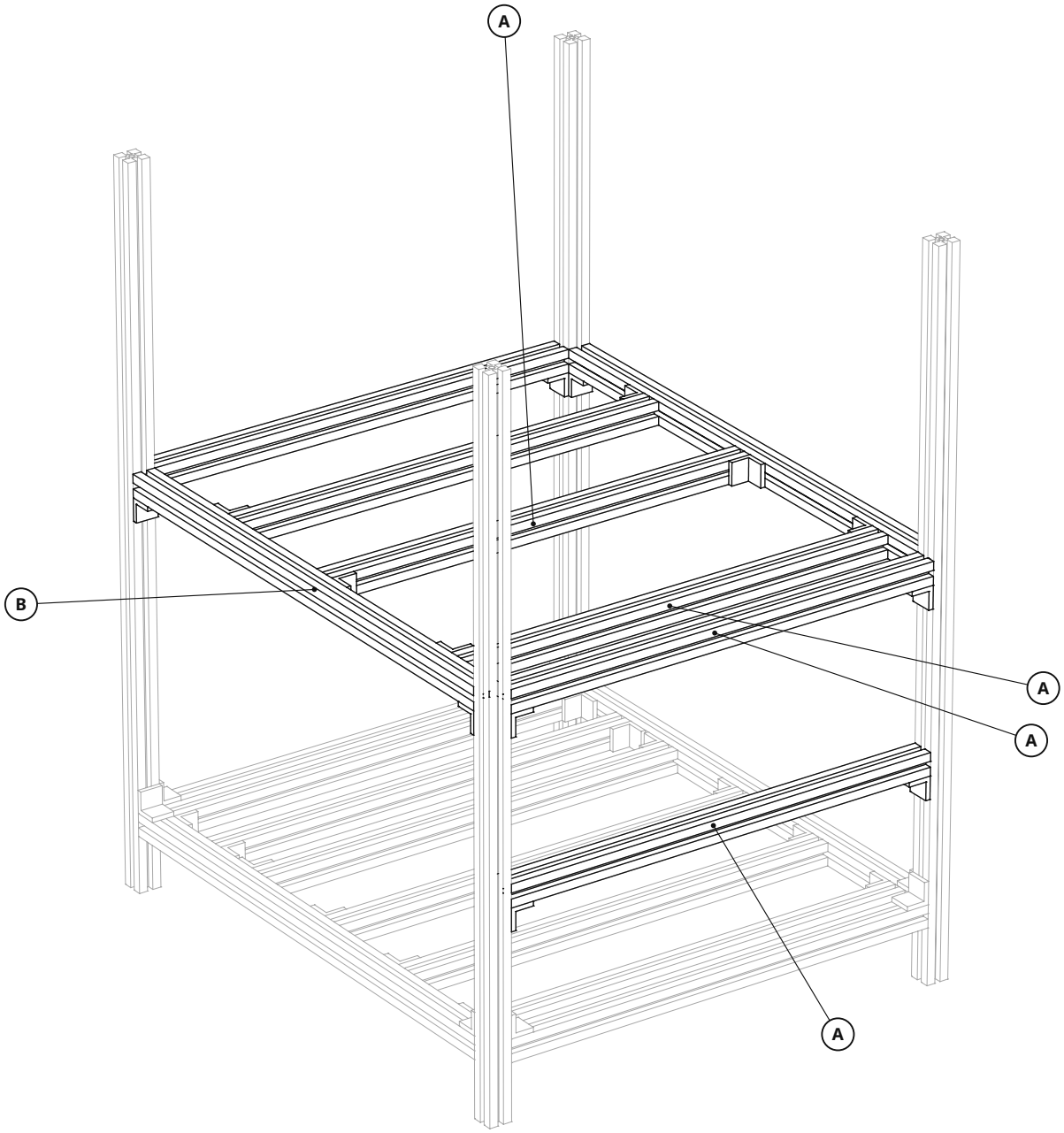
C	4× Extrusion 621 mm
K	4× End cap



! The posts reach 50 mm below the base frame – these are the feet. Fit an end cap at the bottom and top.

5 Mid frame and riser rails

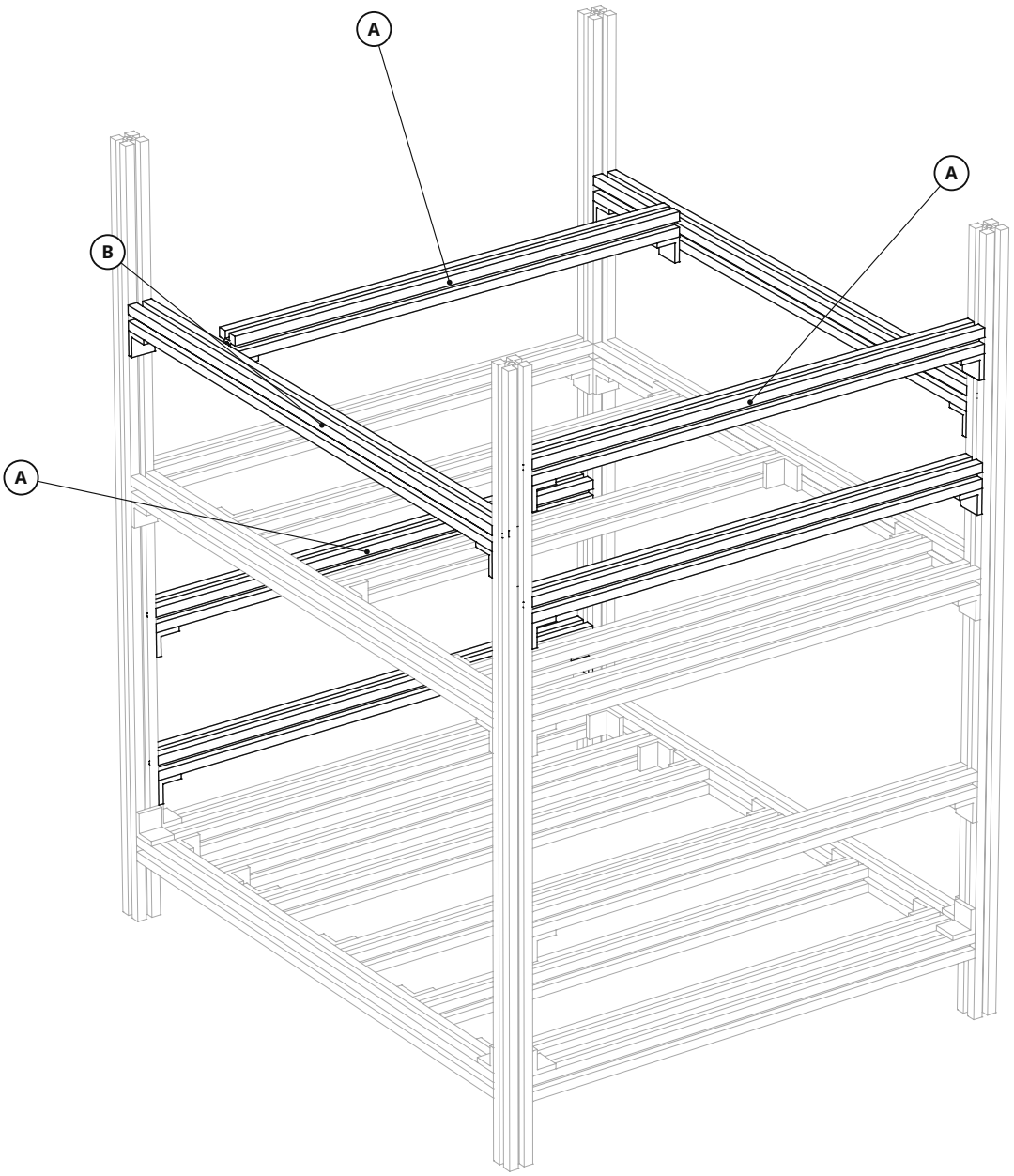
A **6×** Extrusion 390 mm	B **2×** Extrusion 460 mm	W **16×** Bracket + M4 set



! Top of the mid frame at Z = 311 mm (floor to top edge). Riser rails and GPU support run in X.

6 GPU level

B **2×** Extrusion 460 mm	A **5×** Extrusion 390 mm	W **14×** Bracket + M4 set



! The front fan rails carry the three front fans later – do not forget the T-nuts.

7 Top frame

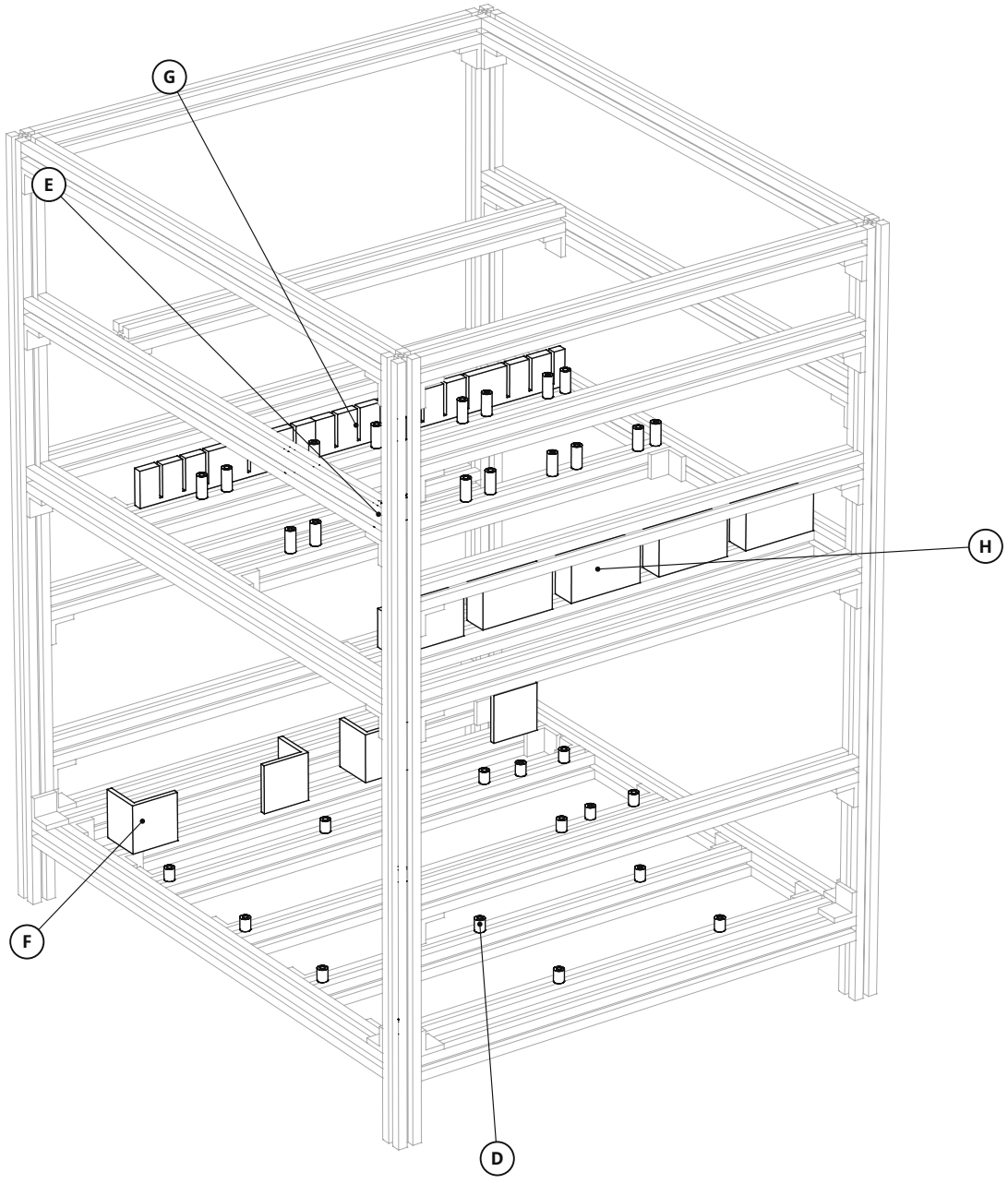
- (A) **2×** Extrusion 390 mm
- (B) **2×** Extrusion 460 mm
- (W) **8×** Bracket + M4 set
- (K) **4×** End cap



! Fit the remaining four end caps on top of the posts.

8 Mount the printed parts

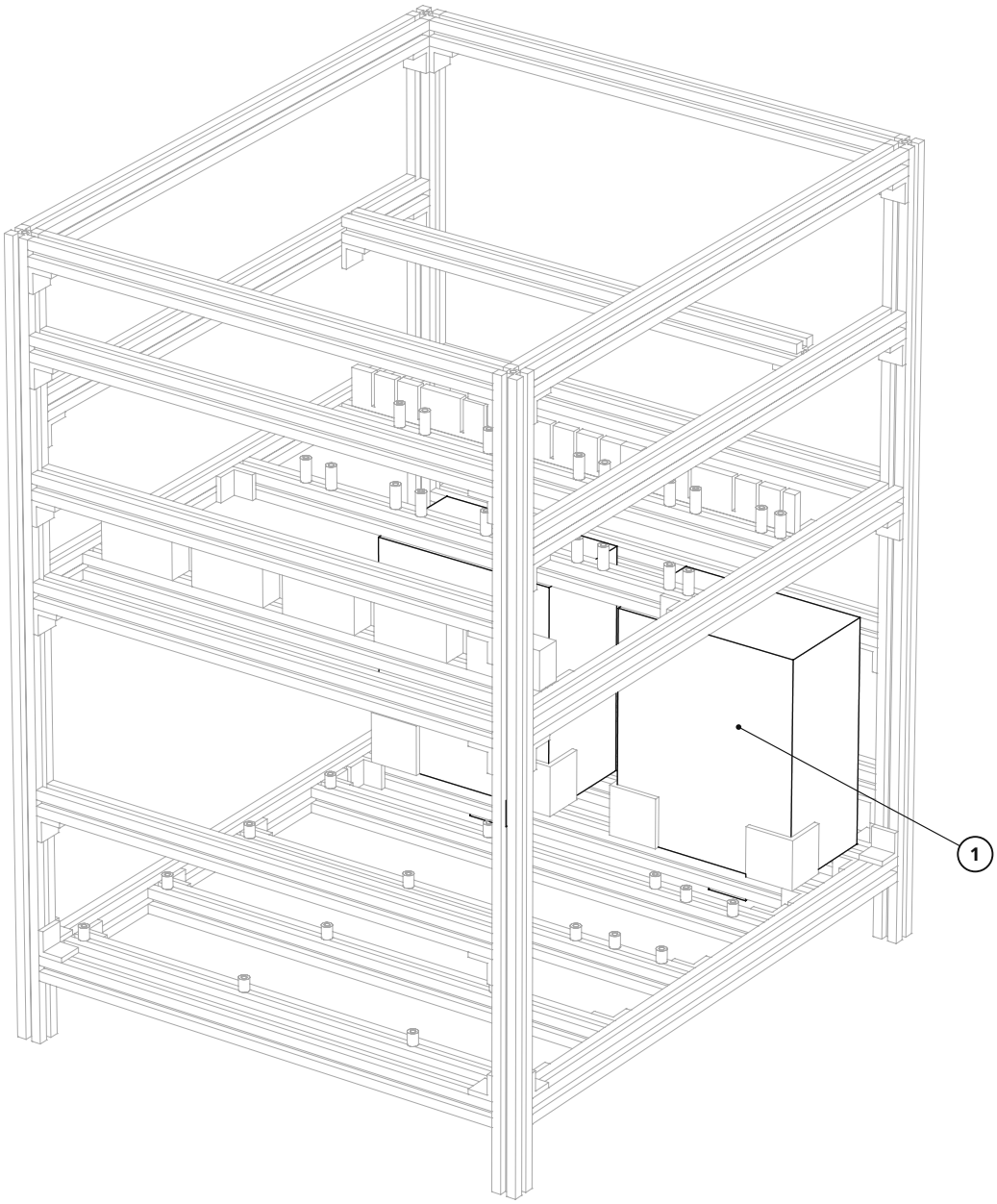
- (D) **16×** Standoff 8×10
- (F) **4×** PSU holder
- (E) **20×** Standoff 8×18
- (H) **5×** GPU rest
- (G) **3×** Bracket comb
- (M4) **24×** M4×12



! Standoffs, PSU holders, comb and GPU rests go onto the pre-loaded T-nuts. How the standoffs are fixed to the extrusion is still a placeholder in the model (M4 from below / glued) – decide before printing.

9 Power supplies

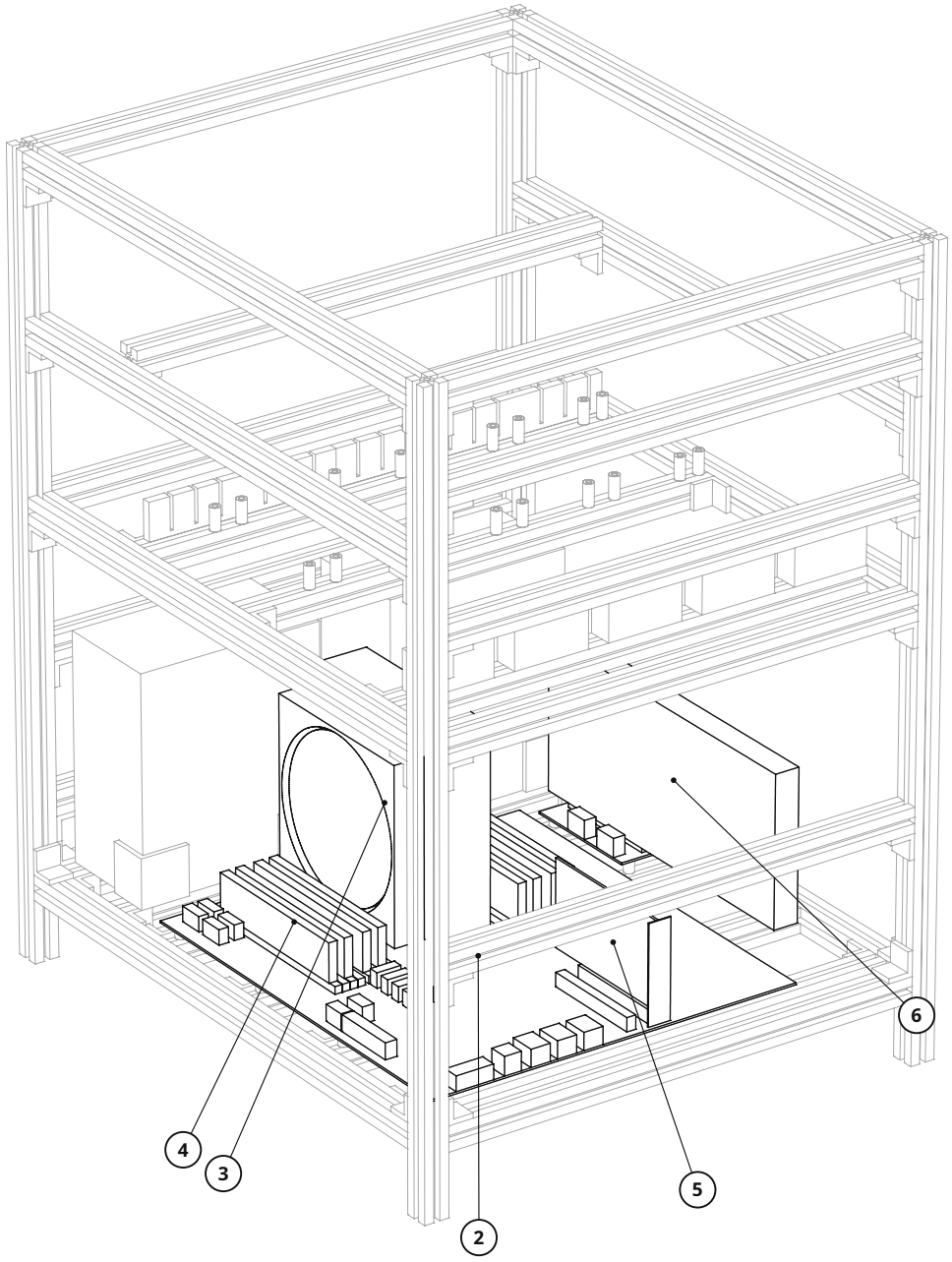
(1) **2×** ZEUS 1650R	(P) **2×** IEC plug



! Insert upright: power inlet down, fan to the rear, modular sockets on top. PSU 1 left, PSU 2 right.

10 Mainboard, CPU, cooler, RAM

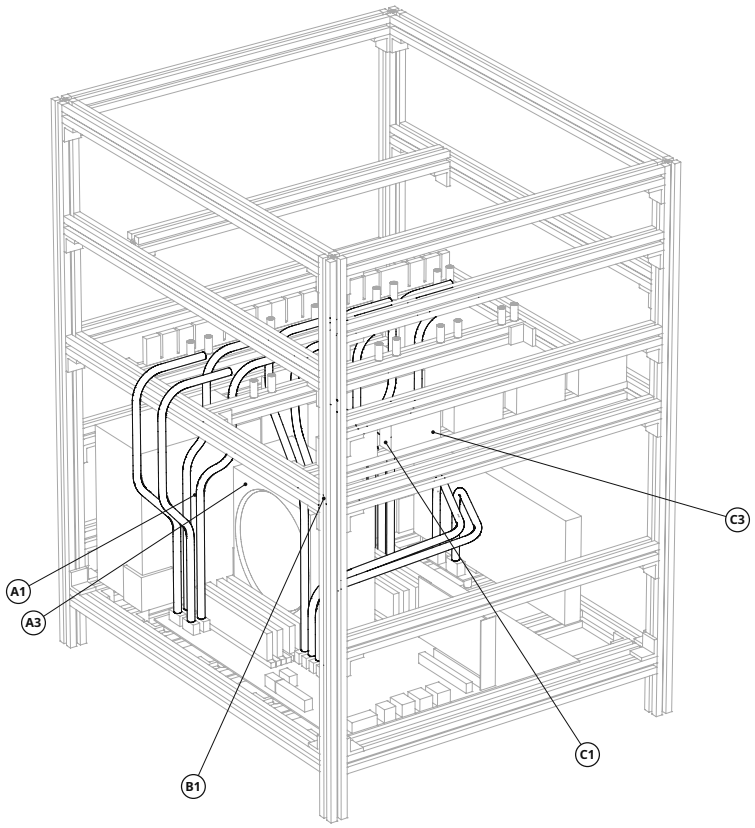
- (2) **1×** Mainboard
- (6) **1×** M.2 riser
- (6) **1×** Quad M.2
- (3) **1×** NH-U14S
- (4) **8×** RDIMM
- (5) **1×** NIC 40G



! Board rotated 180° (I/O to the front). Cooler rotated 90°: airflow to the right (+X). RDIMMs in the outer slot of each channel. NIC in PCIE8. Wear the ESD strap!

11 SlimSAS cables on the board

S **12×** SlimSAS 8i 500

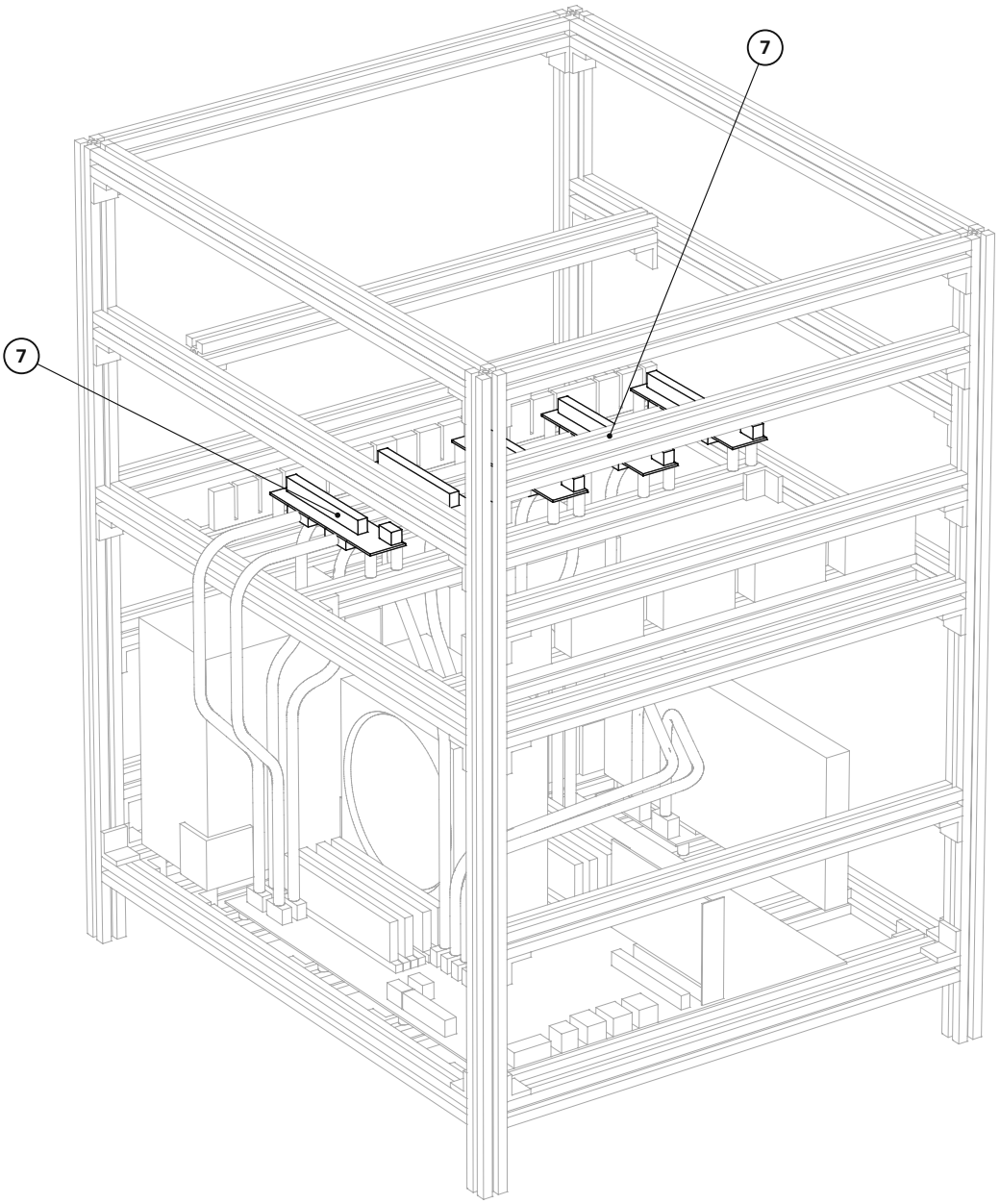


! Plug in straight, do not bend the first 45 mm, bend radius ≥ 25 mm. A1/A2 run diagonally out of the frame to the left, up the outside and back over the mid frame to GPU1.

Port	Target	Route mm	Spare mm
A1	GPU1 · riser socket 1	372	128
A2	GPU1 · riser socket 2	369	131
A3	GPU2 · riser socket 1	357	143
A4	GPU2 · riser socket 2	353	147
B1	GPU3 · riser socket 1	453	47
B4	GPU3 · riser socket 2	460	40
C1	GPU4 · riser socket 1	357	143
C2	GPU4 · riser socket 2	357	143
C3	GPU5 · riser socket 1	356	144
C4	GPU5 · riser socket 2	356	144
B2	M.2 riser socket 1 (quad M.2 option)	453	47
B3	M.2 riser socket 2 (quad M.2 option)	464	36
B2	GPU6 · riser socket 1 (6th GPU option)	411	89
B3	GPU6 · riser socket 2 (6th GPU option)	434	66

12 Fit the risers

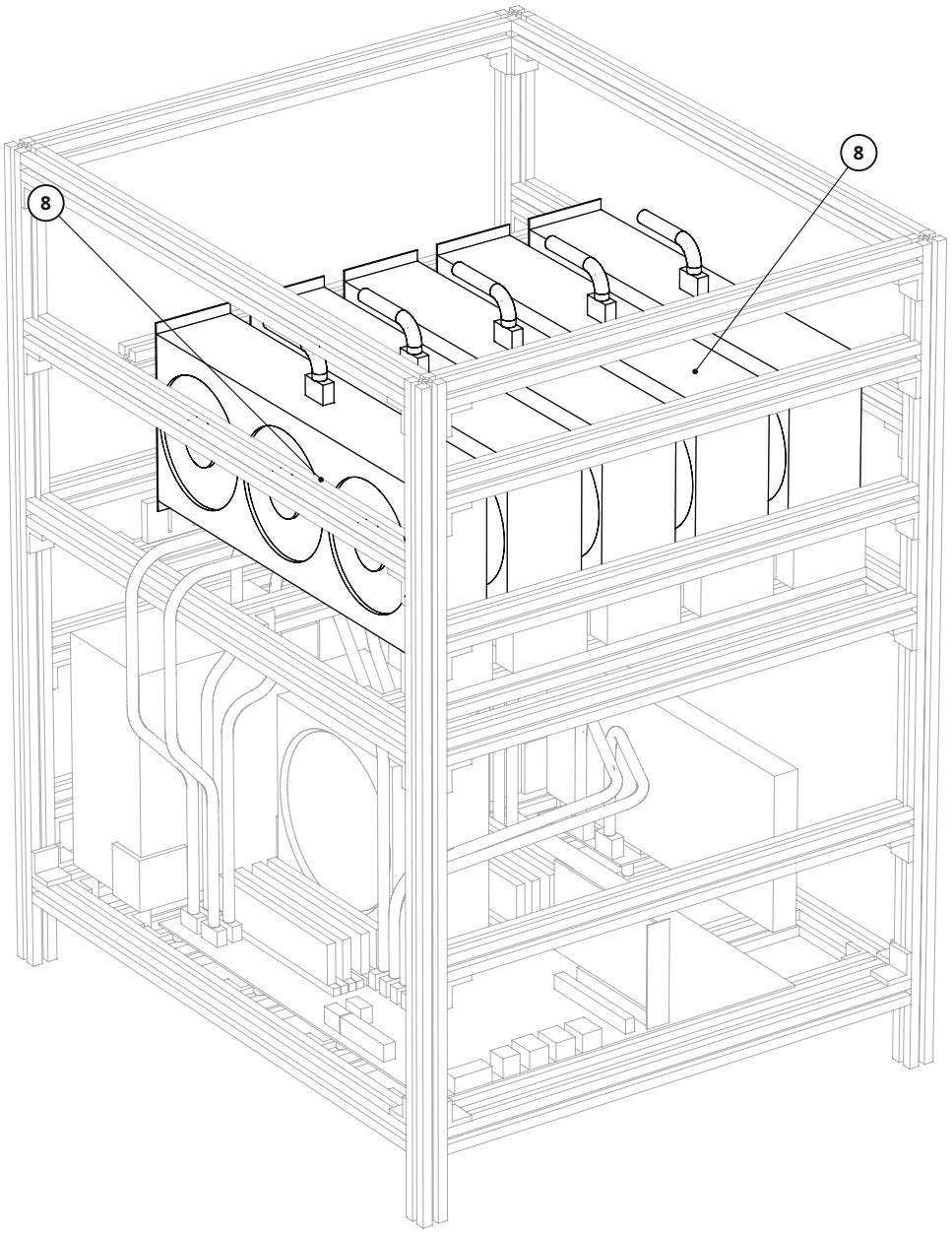
7 **5×** C-Payne riser



! C-Payne risers with the SlimSAS sockets facing left, onto the standoffs (M3). Plug the cables in horizontally from the left.

13 Graphics cards

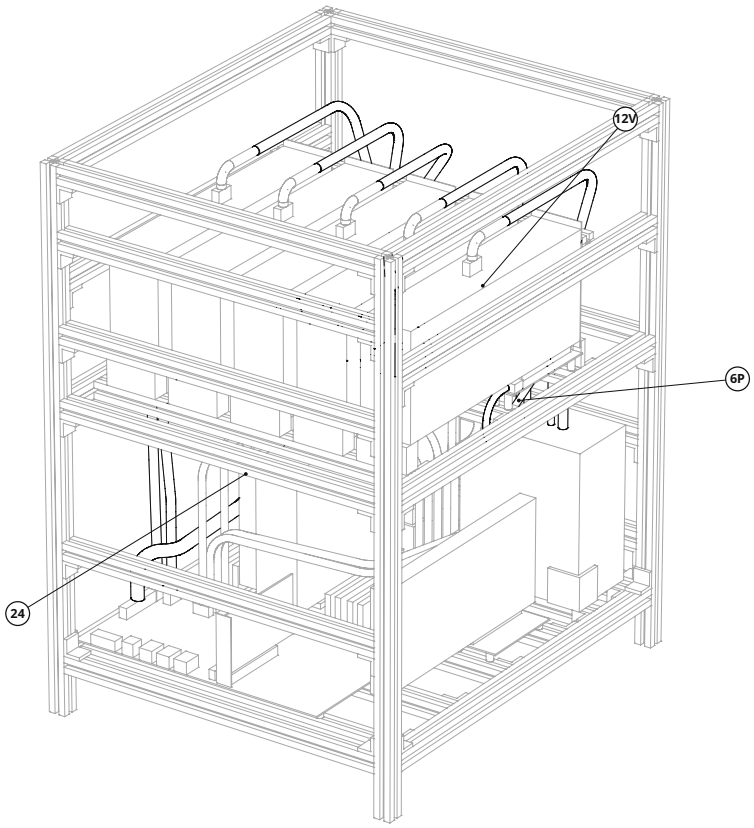
8 **5×** RTX 4090



! All cards the same way: fans to the left, bracket at the rear in the comb, front end on the GPU rest. 12V-2x6 connector on top.

14 Power cables

P **13×** PSU cable



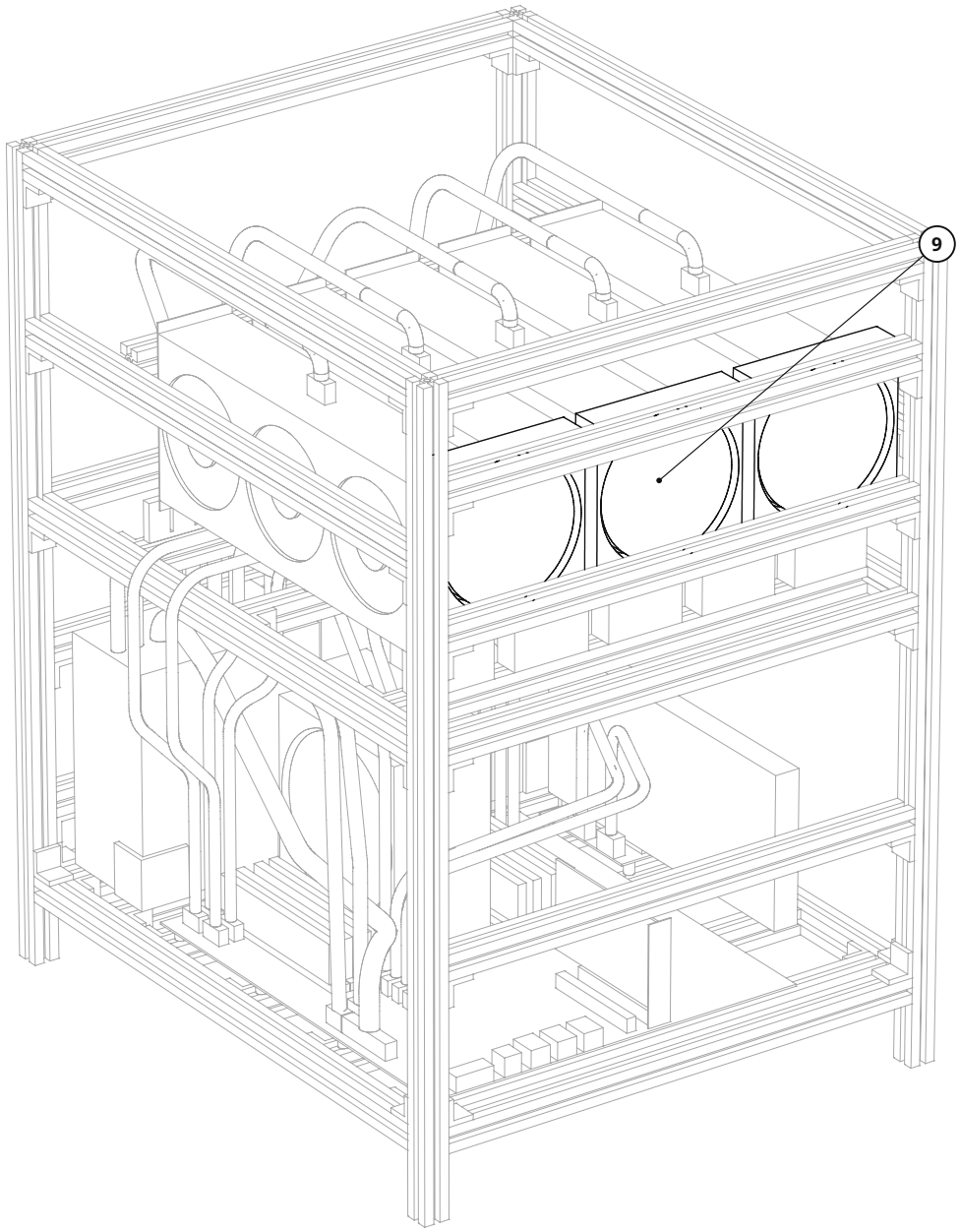
! PSU 1 feeds board, CPU and GPU1–3, PSU 2 feeds GPU4–6. Push 12V-2x6 plugs fully home (click), do not bend them sideways.

PSU	Cable	Target	Route mm	Cable mm	Spare
PSU 1	24-Pin	24-pin mainboard	434	600	166
PSU 1	EPS 4+4	EPS CPU 1	578	750	172
PSU 1	EPS 4+4	EPS CPU 2	579	750	171
PSU 1	12V-2x6	GPU1 (12V-2x6)	463	600	137
PSU 1	12V-2x6	GPU2 (12V-2x6)	443	600	157
PSU 1	12V-2x6	GPU3 (12V-2x6)	452	600	148
PSU 2	12V-2x6	GPU4 (12V-2x6)	443	600	157
PSU 2	12V-2x6	GPU5 (12V-2x6)	461	600	139
PSU 2	12V-2x6	GPU6 (12V-2x6) · optional	246	600	354
PSU 1	PCIe 6+2	riser 1 (6-Pin)	277	600	323
PSU 1	PCIe 6+2	riser 2 (6-Pin)	277	600	323
PSU 1	PCIe 6+2	riser 3 (6-Pin)	301	600	299
PSU 2	PCIe 6+2	riser 4 (6-Pin)	282	600	318
PSU 2	PCIe 6+2	riser 5 (6-Pin)	382	600	218

15 Front fans

9 **3×** Fan 120

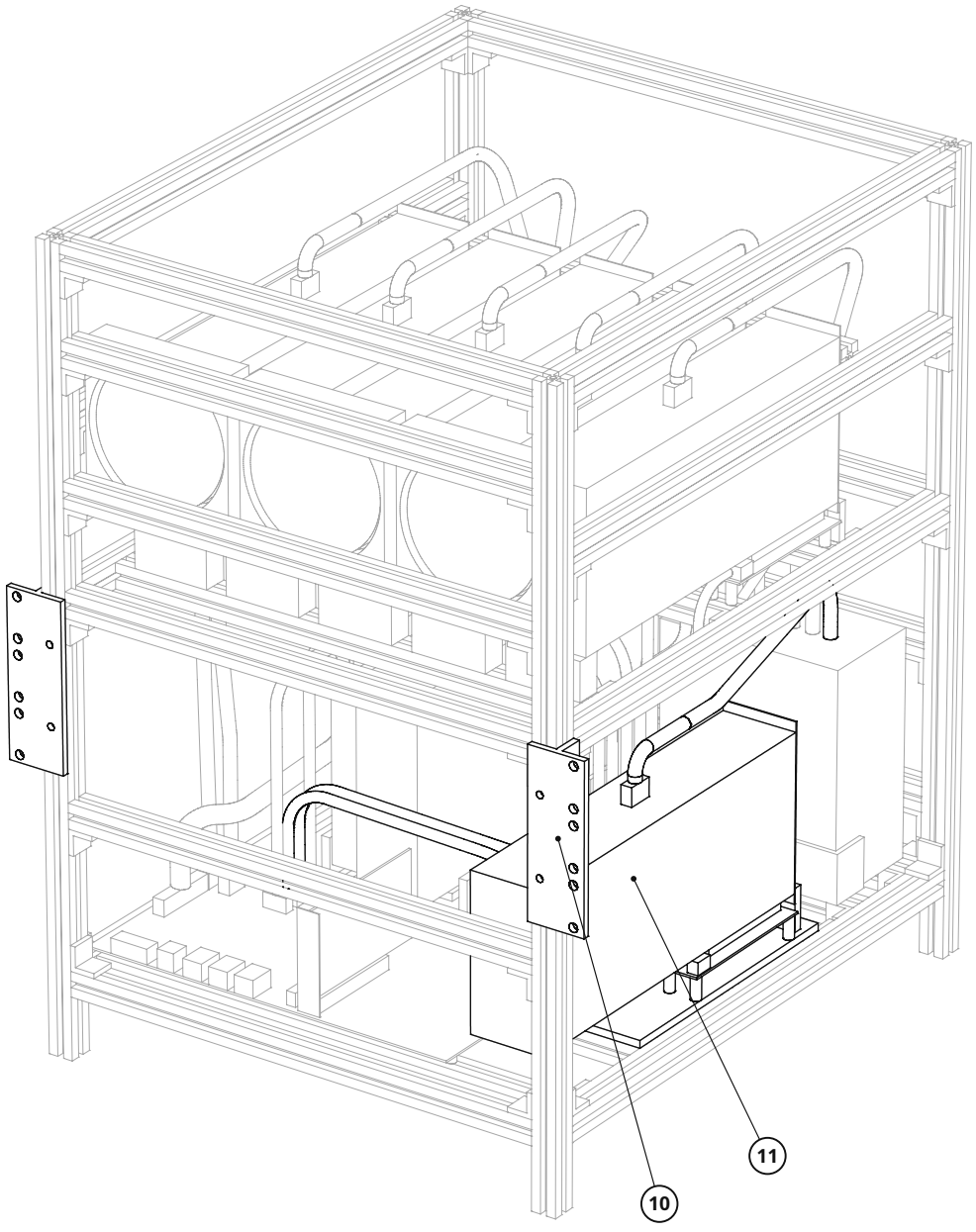
M4 **12×** M4×30



! Airflow to the rear (frame arrow pointing inwards).

16 Options

- (10) **2×** Rack ear
- (8) **1×** RTX 4090
- (7) **1×** C-Payne riser
- (11) **1×** GPU6 holder
- (E) **4×** Standoff 8×18
- (S) **2×** SlimSAS 8i 500
- (P) **1×** PSU cable
- (M4) **8×** M4×10



! Rack ears are retention only – the case stands on rails or a shelf. 6th GPU: remove the quad M.2 card and its riser, move B2/B3 to the GPU6 riser.

Check before building placeholders in the model

SlimSAS plug under the neighbouring riser
After its 45 mm rigid plug, each cable passes just under the edge of the riser to its left (0.5 mm overlap in the model; risers are 47.8 mm apart). Check with the real plug before tightening; if needed raise the neighbouring riser by 2–3 mm (8×20 standoffs).

Left-side overhang
The two GPU1 cables (A1/A2) stick out about 15 mm on the left. In a 19" rack (450 mm clear width) shift the case slightly to the right.

Standoff fixing
How the printed standoffs are held on the T-nut (screw from below, pressed-in nut or glue) is not designed yet.

Riser dimensions
Position of the 6-pin socket and the mounting holes on the C-Payne riser are placeholders. Riser 5 overhangs the right inner frame edge by 4 mm.

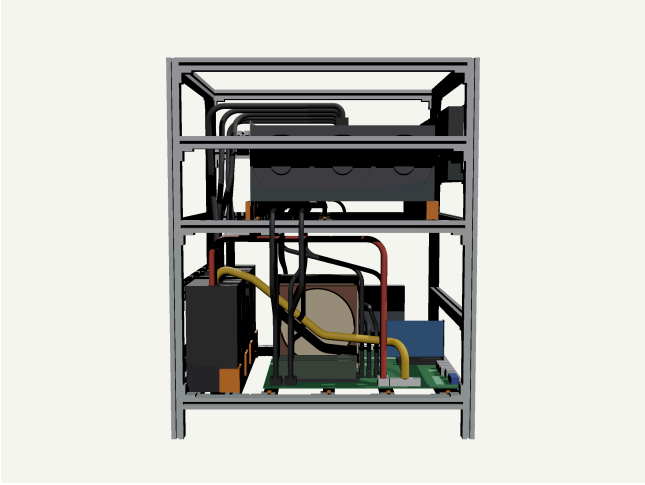
PSU sockets
Per ZEUS 1650R in use at the same time: PSU 1 24-pin + 2× EPS + 3× 12V-2x6 + 3× PCIe; PSU 2 3× 12V-2x6 + 2× PCIe. Check the modular socket layout in the manual (the 12V-2x6 adapter cable uses two sockets).

PSU 1 load
PSU 1 carries CPU, board and three RTX 4090 (up to approx. 1750 W peak). Set a GPU power limit of approx. 350–400 W.

Done.







Bill of materials from the CAD model

Qty	Part number	Description
21	AL-2020-I5-390	Aluminium extrusion 20x20 I-type slot 5, 390 mm
8	AL-2020-I5-460	Aluminium extrusion 20x20 I-type slot 5, 460 mm
4	AL-2020-I5-621	Aluminium extrusion 20x20 I-type slot 5, 621 mm (571 + 50 mm foot)
1	ASRock-ROMED16QM3	ASRock Rack ROMED16QM3 (SP3, SSI-EEB), mounted rotated 180°
6	CPAYNE-X16-BOTTOM	C-Payne PCIe device adapter x8/x16 BOTTOM rev. 2, 2× SlimSAS 8i, 125×30 (1 optional)
16	PRINT-STANDOFF-8x10	Standoff Ø8×10, M3 heat-set insert on top (PETG)
24	PRINT-STANDOFF-8x18	Riser standoff Ø8×18, M3 heat-set insert on top (PETG)
5	PRINT-GPU-REST	GPU front rest, 61 mm wide (PETG)
1	PRINT-GPU6-HOLDER	6th GPU holder: riser plate + bracket stop (PETG, optional)
1	PRINT-COMB-1	Bracket comb segment 1, GPU1–2 (PETG)
1	PRINT-COMB-2	Bracket comb segment 2, GPU3 (PETG)
1	PRINT-COMB-3	Bracket comb segment 3, GPU4–5 (PETG)
1	PRINT-PSU-HOLDER-1A	PSU holder, PSU 1 outer (PETG)
1	PRINT-PSU-HOLDER-1I	PSU holder, PSU 1 inner (PETG)
1	PRINT-PSU-HOLDER-2A	PSU holder, PSU 2 outer (PETG)
1	PRINT-PSU-HOLDER-2I	PSU holder, PSU 2 inner (PETG)
1	PRINT-RACK-EAR-L	19" rack ear left, retention only (PETG, optional)
1	PRINT-RACK-EAR-R	19" rack ear right, retention only (PETG, optional)
3	FAN-120x25	Fan 120×120×25
6	GPU-RTX4090-3S	NVIDIA RTX 4090, 3-slot, 305 mm (1 optional as 6th GPU)
2	IEC-C13-PLUG	IEC C13 power cord plug (placeholder)
1	NIC-40G	40G NIC, low profile, in PCIE8 (placeholder)
1	Noctua-NH-U14S-TR4-SP3	Noctua NH-U14S TR4-SP3, rotated 90°, airflow to the right
6	PSU-CABLE-12V2X6	12V-2x6 cable from PSU (1 optional)
5	PSU-CABLE-PCIE	PCIe 6+2 cable to riser
1	PSU-CABLE-ATX24	24-pin ATX cable
1	PSU-CABLE-EPS1	EPS 4+4 CPU cable 1
1	PSU-CABLE-EPS2	EPS 4+4 CPU cable 2
1	QUAD-M2	Quad M.2 card (placeholder, alternative to 6th GPU)
8	RDIMM-DDR4	DDR4 RDIMM, outer slot of each channel
1	RISER-M2	Riser board for the quad M.2 card (placeholder)
14	SLIMSAS-8I-500	SlimSAS 8i (SFF-8654) cable 500 mm, round (10 for GPU1–5, 2 quad M.2, 2 optional GPU6)
2	SST-ZU1650R	Silverstone ZEUS 1650R Titanium, upright, inlet down, fan to the rear
58	BRACKET-2020-I5	Corner bracket 20x20x20, I-type slot 5, with M4 screw + T-nut set

Frame hardware example supplier: Dold Mechatronik (DE), article numbers

Qty	Art. no.	Item	Length	Note
4	62002-Z	Aluminium extrusion 20x20 light, I-type slot 5, cut to length	621 mm	corner posts
21	62002-Z	Aluminium extrusion 20x20 light, I-type slot 5, cut to length	390 mm	all X members
8	62002-Z	Aluminium extrusion 20x20 light, I-type slot 5, cut to length	460 mm	all Y members
58	61951-BS	Bracket 20 I-type slot 5 incl. M4 set (2× M4×8 + 2× T-nut)		every joint
8	61302-1	End cap 20 I-type slot 5		post ends
86	61326-VK	T-nut with web I-type slot 5, M4	M4	standoffs, holders, fans, rack ears (+ spares)
12	DIN7984-4x30	DIN 7984 low-head socket screw, 8.8 zinc	M4x30	front fans
26	DIN7984-4x12	DIN 7984 low-head socket screw, 8.8 zinc	M4x12	printed holders
8	DIN7984-4x10	DIN 7984 low-head socket screw, 8.8 zinc	M4x10	rack ears (optional)



DIY AI GPU SERVER

Open hardware · CERN-OHL-P v2

14U · 2026-10-29